

YAO XIAO

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EDUCATION

Georgia Institute of Technology

Atlanta, GA

Bachelor of Science in Computer Science, Minor in Financial Technologies

Aug 2023 – May 2027

- Threads: **Artificial Intelligence and Modeling & Simulation**
- Honors: **Dean's List** (all eligible semesters), Fall 2023 – Present
- Relevant Coursework: Machine Learning, Deep Learning, Design & Analysis of Algorithms, Computer Organization & Programming, Linear Algebra, Probability & Statistics, Combinatorics

RESEARCH EXPERIENCE

ML and Modeling Researcher

Jan 2026 – Present

Responsible AI for Decision Making – Georgia Tech VIP Program

Advisor: Prof. Pascal Van Hentenryck

- Established optimization models using Mixed Integer Programming and DC Optimal Power Flow for topology switching in power grids; induced congestion by scaling loads (1.09x–1.20x) to identify switching candidates
- Performed dual variable and slack analysis using Gurobi on 14, 30, and 57-bus systems; identified bottleneck constraints (e.g., thermal limit on line $e(8,9)$ with dual of $-9.79\$/MW$)
- Built **Grid Viz** visualization system to map flow directions, capacities, and angle differences
- Leading ML phase: training GNN and DNN models on 14-bus system to predict optimal topology switching, replacing traditional optimization solvers; developing dataset generation with randomized load values

Machine Learning Researcher

Jan 2025 – Aug 2025

Surrogate Modeling for Urban Regeneration Lab (SMUR)

Advisor: Prof. Patrick Kastner

- Conducting research in ML-driven surrogate modeling to accelerate environmental simulations for urban regeneration projects across U.S. cities
- Implementing Pedestrian Environment Index (PEI) methodology: computing walkability subindices using Census API, Overpass API, Python (pandas, NumPy, GeoPandas); building web apps with React, Leaflet, and AWS S3

Digital Real-world Modeling & Ecological Sensitivity Analysis

Aug 2024 – Dec 2025

Context Capture, ArcGIS, InVEST, Python (pandas/NumPy/SALib)

- Used Phantom 4 Pro V2.0 for UAV tilt photography and Context Capture to construct 3D model; performed ecological sensitivity analysis using ArcGIS, integrated with the 3D model

WORK EXPERIENCE

Undergraduate Teaching Assistant

Jan 2025 – Present

PHYS 2211 (Intro Physics I), School of Physics, Georgia Tech

Supervisor: Dr. Emily Alicea-Muñoz

- Lead and facilitate lab sessions; provide one-on-one support to ~90 students per week on mechanics and kinematics

Research and Development Assistant

Jun 2024 – Jul 2024

WeiSheng & Wecom

- Developed AI model functionality; built back-end architecture and performed defect resolution. Contributed to technical routing design and UI design in Agile workflow

SELECTED PROJECTS

Financial Forecasting with LSTM and NLP | *Python, PyTorch, LSTM, FinBERT, BERTopic*

- Built LSTM forecasting models on S&P 500 and NYSE datasets for next-day stock price predictions tracked through MSE/MAE metrics
- Applied NLP methods (FinBERT, Hugging Face, BERTopic) to financial news, linking sentiment and industry signals to market movements for more accurate decision support

Glimpse3D | *Python, JavaScript, Three.js, TripoSR, PyTorch, Computer Vision*

- Developed real-time 3D mesh generation from a single webcam image using TripoSR for single-view 3D reconstruction, rembg/U2Net for background removal, and Three.js for interactive rendering

TECHNICAL SKILLS

Languages: C++, C, Java, Python, JavaScript, TypeScript, Terraform, HTML, CSS, SQL, MATLAB, Assembly

Frameworks: Django, ReactJS, Spring Boot, JUnit, Redux

ML/AI: PyTorch, FinBERT, Hugging Face, BERTopic, LSTM, GNN, DNN, TripoSR, rembg/U2Net

Tools & Libraries: Docker, Kubernetes, Git, Jupyter, Gurobi, SciPy, NumPy, pandas, GeoPandas, Leaflet, AWS S3

Domain: ArcGIS, InVEST, Context Capture, Overpass API, Census API

LEADERSHIP & ACTIVITIES

Squad Leader, Yellow Jacket Fencing Club – Men's Sabre

Aug 2023 – Present

Georgia Institute of Technology

- Manage and organize weekly training for Sabre team; provide competition instruction and team coaching
- Helped Georgia Tech win third place in the nation at USACFC 2024